2(a)	**M1** Al_2O_3 is ionic **M2** P_4O_{10} is a (simple) molecule **M3** ionic bonds are stronger than intermolecular forces	**3**
2(b)(i)	$mol^{-2}\ dm^6$	**1**
2(b)(ii)	**M1** at equilibrium: moles of CO = 0.120 **AND** moles of H_2 = 0.240 **M2** $[CH_3OH]$ = 0.056 **AND** [CO] = 0.024 **AND** $[H_2]$ = 0.048 **M3** use of K_c with correct evaluation = 1.01×10^3	**3**
2(b)(iii)	no effect **AND** K_c is not affected by pressure	**1**

Examiner report missing
9701_w25_qp_24 Question 2

1(a)(i)	24 protons 28 neutrons 24 electrons	**1**
1(d)(i)	rate of forward and backward (reaction) are the same / equal **OR** concentrations of reactants and products are constant	**1**
1(d)(ii)	closed system	**1**
1(d)(iii)	**M1** (mixture becomes more) orange **M2** forward reaction is endothermic **M3** (mixture becomes more) orange **M4** equilibrium moves (forwards/right) to compensate for extra acid / H^+	**4**

Examiner report missing
9701_w25_qp_23 Question 1

2(a)(i)	delocalised electrons move (through the structure / between layers)	**1**
2(e)(i)	rate of forward and backward (reaction) are the same / equal **OR** (ratio of) concentrations of reactants and products are constant	**1**
2(e)(ii)	closed system	**1**
2(e)(iii)	**M1** (mixture / solution becomes more) yellow **M2** (forward) reaction is exothermic (equilibrium shifts left) **OR** equilibrium / reaction shifts (left) to absorb (additional) heat / energy **M3** (mixture / solution becomes more) red **M4** fewer moles of gas on right-hand side **OR** (gas on left and) no gas on the right	**4**

Examiner report missing
9701_w25_qp_22 Question 2

1(a)(i)	24 protons 28 neutrons 24 electrons	**1**
1(d)(i)	rate of forward and backward (reaction) are the same / equal **OR** concentrations of reactants and products are constant	**1**
1(d)(ii)	closed system	**1**
1(d)(iii)	**M1** (mixture becomes more) orange **M2** forward reaction is endothermic **M3** (mixture becomes more) orange **M4** equilibrium moves (forwards/right) to compensate for extra acid / H^+	**4**

Examiner report missing
9701_w25_qp_21 Question 1

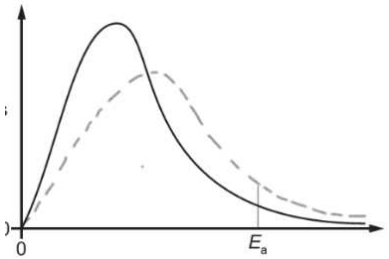
3(a)(i)	0.00658(0) mol	**1**
3(b)	**M1** $0.2 - 0.154 = 0.046$ (mol H_2 and / or I_2 reacts) **M2** mol H_2 at equilibrium $0.1 - 0.046 = 0.054$ **M3** mol HI at equilibrium $0.3 + (0.046 \times 2) = 0.392$	**3**
3(c)	2.46	**1**
3(d)(i)	(electrostatic) attraction between the nuclei (of two atoms) and a shared pair of electrons	**1**
3(d)(ii)	**M1** HCl is more stable **AND** (because) Cl atom is smaller / less distance between electron pair and nucleus / shorter (H-X) bond length **M2** (In HCl) stronger attraction between nucleus and shared pair of electrons **OR** (In HCl) (fewer shells of electrons so) less shielding **OR** (In HCl) greater orbital overlap so stronger interaction	**2**
3(d)(iii)	any value or range of values for K_c between $5.50 \times 10^{-34} < K_c < 1.26 \times 10^{-3}$	**1**

Examiner report missing
9701_s25_qp_24 Question 3

2(a)	Na_2O	**1**

2(d)	it / ZnO accepts protons	**1**

Examiner report missing
9701_s25_qp_23 Question 2

3(a)(i)	 *Two correct for one mark, all four for two marks:* • start at origin • peak sits below the original • peak sits to the right-hand side of the original • new line crosses the original once only **AND** to the right of the original peak	**2**

3(b)	**M1** (if a change is made at a system is dynamic equilibrium) the equilibrium moves **M2** to minimise the change in conditions	**2**

3(c)	(forward) reaction is exothermic $(a + b) > c$	**2**

Examiner report missing
9701_s25_qp_21 Question 3

1(a)(i)	• (+)3 / III • –1 / I • (+)5 / V • –1 / I	**2**

1(a)(ii)	high pressure low temperature	**2**

Examiner report missing
9701_m25_qp_22 Question 1

3(a)(i)	bromine: (dark / red-)brown liquid iodine: grey / black solid	**1**

3(c)(i)	sodium iodide / NaI	**1**

3(c)(ii)	sodium chloride / NaC*l*	**1**

Examiner report missing
9701_m25_qp_22 Question 3

3(a)(i)		**3**

element	C (graphite)	Sn
state and appearance	grey shiny solid	silvery solid
electrical conductivity	● good/ conductor	good
type of bonding	● covalent	metallic
type of structure	giant	● giant

3(b)(i)	$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$	**1**

3(b)(ii)	thermal stability **increases** down the group	**1**

3(b)(iii)	**M1** NH_4^+ / ammonium ion / ammonium cation is a proton / $\mathbf{H^+}$ donor **M2** OH^- is a proton / $\mathbf{H^+}$ acceptor **OR** the OH^- / NaOH is a stronger base than NH_3	**2**

Question 3

(a) (i) This question was well answered with many candidates stating that graphite was a conductor of electricity with covalent bonding and that tin was a giant structure.

(ii) Many candidate answers described the layers/hexagon rings with IMFs between layers, a feature of graphite that was not required in this question.

(iii) Many candidates' answers did not gain any credit for either not mentioning that the electrons were 'delocalised' or that they were 'moving', which are the necessary criteria for tin to conduct electricity.

(b) (i) The most common errors were to give the formula for magnesium chloride as MgCl and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

(ii) Increases was correctly stated by many candidates.

(iii) Many candidates gained some credit for appreciating that the ammonium carbonate donated a proton and the NaOH accepted this proton. Few answers gained full credit for stating that the ammonium ion, NH_4^+, is the actual species that behaves as an acid, by donating the proton, and it is the OH– ion that is accepting these protons.

(c) (i) Correct completion of the graph was shown by many candidates.

(ii) Incorrect answers often omitted state symbols or gave unbalanced equation with $2e^-$ on the right hand side of the equation.

(d) (i) There were many correct candidate answers stating an increase in the number of electrons, from CH_4 to PbH_4, often followed by incorrect comments that there was an increased nuclear attraction between the nucleus and outer electrons, or that the nuclear charge outweighed the shielding effect. After discussing the impact of the increased intermolecular forces of attraction, IMFs, down the group, many candidates went on to discuss the breaking of bonds.

(ii) The shape of a molecule of SiH_4 was well answered.

(e) (i) Many correct answers for the equation were present. A common error was to use HCl as a reactant, instead of Cl_2.

(ii) Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when SiCl_4 is added to water.

(iii) Credit was given to candidates for recognising that the structure and bonding of solid, SiO_2, was giant covalent and liquid SiCl_4 as simple molecular/simple covalent. There were few correct explanations relating to the difference in melting points of these two compounds. Many candidates referred to the need for more energy to break the strong covalent bonds in SiO_2 but failed to mention anything regarding the energy requirements for breaking the intermolecular forces of attraction, IMFs, between liquid SiCl_4 molecules. Several answers compared the strength of the covalent bonds in both SiO_2 and SiCl_4 and many answers incorrectly stated that the structure of SiO_2 contained double bonds between the Si and O atoms.

(f) Many candidates gave the correct answer.

3(a)(i)	(natural =)lightning	**1**
	(man-made =)internal combustion engines	**1**
3(b)(i)	(+)5 / V	**1**
3(b)(iii)	(NH_3 is an) H^+ acceptor	**1**

Question 3

(a) (i) Many candidate answers referred to lightning as responsible for one natural occurrence of NO_2. A common incorrect answer was acid rain. Fewer candidate answers described the main source of man-made nitrogen oxides as the internal combustion engine. Lots of candidates incorrectly identified the source as catalytic converters or 'car exhaust' as the origin of nitrogen oxides.

(ii) Very few candidates gave appropriate species in the equation and if they did these equations were not always balanced correctly. Many missed the significance of the word 'direct' in the question and attempted to give an equation showing the role of NO_2 as a catalyst in the production of sulfur-based acid rain.

(iii) Candidates who were familiar with this area of the syllabus described how PAN forms from NO_2. Common incorrect descriptions included acid rain, water, oxygen, carbon monoxide, carbon dioxide or ozone reacting with NO_2.

(iv) Correctly balanced equations were common. Typical errors included incorrect formula of nitric acid or calcium nitrate or production of H_2 rather than H_2O.

(v) Some candidate answers described the production of a brown gas. Others focused on the production of a white solid even though the original compound is also a white solid.

(b) (i) Occasionally incorrect oxidation states were seen; –1 and +3 were common incorrect answers.

(ii) The majority of candidates identified aluminium as the correct species that is oxidised. The weakest answers described one of the two products of the reaction.

(iii) Excellent answers gave a full description of the formation of a dative bond due to the lone pair on the nitrogen atom accepting a proton. Weaker candidate descriptions only referred to the ability of ammonia to form a dative bond.

(iv) Identification of an appropriate shape for $Al(OH)_4^-$ was present in a significant proportion of answers.

(c) (i) Inclusion of state symbols for ionisation energy equations are essential. Some equations showed addition of an electron, other equations were incorrectly balanced.

(ii) A general increase in ionisation energy was shown by most candidates. Many answers showed a larger increase in energy for one of the ionisation processes, with the exact ionisation process often incorrectly identified. Weaker answers confused the trend in ionisation energy across a period with the trend of successive ionisation energies.

(d) Phosphorus was sometimes identified incorrectly with ionic bonding or as an ionic structure.

(e) Use of the term 'simple' to identify the structure of solids made up of discrete molecules was not well known and the structure of solid nitrogen was often referred to as 'giant'.

(f) (i) This question highlighted many misconceptions about hybridisation, sigma and pi bonds. These include

- a triple bond involves formation of two sigma bonds and one pi bond
- triple bonds are made when sigma and pi overlap
- hybridisation in P_2 identified as sp^2 or sp^3
- in P_2 overlap of s orbitals creates a sigma bond
- formation of sigma bonds by side-on overlap of orbitals and pi bonds by head-on overlap of orbitals.

(ii) Many candidate responses referred to the appropriate difference in bond strength. Some answers were incomplete, with no comparison of reactivity given. Some candidates misinterpreted the question and explained why they had different bond energy values.

3(a)(i)	<table><tr><td>element</td><td>C (graphite)</td><td>Sn</td></tr><tr><td>state and appearance</td><td>grey shiny solid</td><td>silvery solid</td></tr><tr><td>electrical conductivity</td><td>● good/ conductor</td><td>good</td></tr><tr><td>type of bonding</td><td>● covalent</td><td>metallic</td></tr><tr><td>type of structure</td><td>giant</td><td>● giant</td></tr></table>	**3**
3(b)(i)	$MgCO_3 + 2HCl \rightarrow MgCl_2 + CO_2 + H_2O$	**1**
3(b)(ii)	thermal stability **increases** down the group	**1**

Question 3

(a) (i) This question was well answered with many candidates stating that graphite was a conductor of electricity with covalent bonding and that tin was a giant structure.

(ii) Many candidate answers described the layers/hexagon rings with IMFs between layers, a feature of graphite that was not required in this question.

(iii) Many candidates' answers did not gain any credit for either not mentioning that the electrons were 'delocalised' or that they were 'moving', which are the necessary criteria for tin to conduct electricity.

(b) (i) The most common errors were to give the formula for magnesium chloride as $MgCl$ and to use 1 mole of hydrochloric acid, in the equation, instead of 2 moles.

(ii) Increases was correctly stated by many candidates.

(iii) Many candidates gained some credit for appreciating that the ammonium carbonate donated a proton and the NaOH accepted this proton. Few answers gained full credit for stating that the ammonium ion, NH_4^+, is the actual species that behaves as an acid, by donating the proton, and it is the OH– ion that is accepting these protons.

(c) (i) Correct completion of the graph was shown by many candidates.

(ii) Incorrect answers often omitted state symbols or gave unbalanced equation with $2e^-$ on the right hand side of the equation.

(d) (i) There were many correct candidate answers stating an increase in the number of electrons, from CH_4 to PbH_4, often followed by incorrect comments that there was an increased nuclear attraction between the nucleus and outer electrons, or that the nuclear charge outweighed the shielding effect. After discussing the impact of the increased intermolecular forces of attraction, IMFs, down the group, many candidates went on to discuss the breaking of bonds.

(ii) The shape of a molecule of SiH_4 was well answered.

(e) (i) Many correct answers for the equation were present. A common error was to use HCl as a reactant, instead of Cl_2.

(ii) Most candidates gave the correct observations of steamy/misty/white fumes or a white precipitate when $SiCl_4$ is added to water.

(iii) Credit was given to candidates for recognising that the structure and bonding of solid, SiO_2, was giant covalent and liquid $SiCl_4$ as simple molecular/simple covalent. There were few correct explanations relating to the difference in melting points of these two compounds. Many candidates referred to the need for more energy to break the strong covalent bonds in SiO_2 but failed to mention anything regarding the energy requirements for breaking the intermolecular forces of attraction, IMFs, between liquid $SiCl_4$ molecules. Several answers compared the strength of the covalent bonds in both SiO_2 and $SiCl_4$ and many answers incorrectly stated that the structure of SiO_2 contained double bonds between the Si and O atoms.

(f) Many candidates gave the correct answer.

2(a)	58% / no change / stays the same **AND** catalysts don't affect yield / iron is a catalyst	**1**
2(b)(i)	less yield (in **C** than **A**) **AND** (as) exothermic (forward reaction)	**1**
2(c)(i)	**M1** $K_p = (pNH_3)^2 / (pN_2)(pH_2)^3$ **M2** atm^{-2}	**2**
2(c)(ii)	**M1** correct amount N_2 at equilibrium = 1 – 1.16 / 2 = 0.42 **AND** correct amount H_2 at equilibrium = 3.0 – (1.16 × 3 / 2) = 1.26 **M2** express amount of ammonia as a mol fraction in terms of total amount of $N_2+H_2+NH_3$ at equilibrium = 1.16 / (N_2 + H_2 + 1.16) = 0.408 **OR** 0.41	**2**
2(c)(iii)	0.625 × 1000 = 625 (atm)	**1**

Question 2

Although **(d)** focused on inorganic chemistry, the majority of this question was based on equilibria and rates. Better performing candidates were able to discern the differing aspects here: rates and collision theory formed one notion; equilibria, yield and Le Chatelier's principle the other.

(a) This was generally well answered, with many candidates identifying the null effect of a catalyst on equilibrium position. The best answers stated this explicitly.

(b)(i) This was well answered by candidates, who could link lower yield with the exothermic reaction. Precision of wording was key here.

(ii) Many candidates attempted to answer this rates question in terms of equilibrium yield and were unable to gain full credit as a result. If an item pairs rate and yield within a question, candidates are advised to read the stem of the question carefully and to give relevant answers.

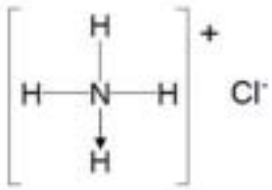
(c)(i) Many candidates were unable sufficiently to distinguish between K_c and K_p. Square brackets must not be used in K_p expressions to avoid confusion with concentration variables. Partial pressure variables can be given as p_X, P_X, $p(X)$, $P(X)$ or $pp(X)$.

(ii) Many numerical answers were correct, though common errors included not adjusting the equilibrium concentrations of N_2 and H_2, or not accounting for **2**NH_3 or **3**H_2 in the stoichiometric equation.

(iii) This was generally well answered by candidates, although some errors of powers of ten were seen in a minority of responses.

(d)(i) Candidates' knowledge of this nitrogen chemistry seemed largely secure, though correct quotation of three equations (two showing NO_2 catalysis and one of the conversion of SO_3 to acid rain) were needed to gain full credit, alongside a statement of the role of NO_x.

(ii) There were many good partial answers to this question, with most referring to PAN but with less precision on how this is produced.

4(a)	 **OR** Correct representation or label showing **M1** (ionic bond between) NH_4^+ and Cl^- **AND** 4 N—H covalent bonds OR 4 shared pairs of electrons in NH_4 **M2** 1 N—H coordinate bond OR arrow on bond/line in correct direction or only 1 bonding pair using the same symbol for electrons1	**2**

4(b)(i)	acid–base reaction	**1**

Question 4

Many individual parts of this question were answered well. Parts **(a)**, **(c)(i)**, **(e)(ii)** and **(f)(ii)** were clearly the most challenging, often as candidates were required to apply several independent parts of their knowledge of the syllabus.

(a) Candidates were not strictly required to give dot-and-cross diagrams, though these and modified Lewis diagrams often produced the clearest responses. Many answers incorrectly showed a pentavalent nitrogen atom — a kind of NH_5 species, with little understanding of the NH_4^+ Cl^- superstructure. The dative bond needed to be clearly distinct from other covalent bonds and ionic charges needed to be given.

(b)(i) There were many correct answers to this question.

(ii) Different versions of this answer were offered, but a large number were a simple addition of NH_4^+ and OH^-, which did not gain credit.

(c)(i) This was well answered by candidates whose knowledge extended to terminology associated with electron-pair donors.

(ii) This was found to be the most challenging item on the paper which required candidates to assess either the basicity or nucleophilicity of each species. Opposing answers were therefore acceptable with enough reasoning.

(d) This was generally well answered by candidates, but the single-step nature of the S_N2 mechanism needed to be clearly evidenced by one transition state; the slower rate of this reaction by a higher energy of the TS relative to either produced in S_N1. Candidates frequently accessed the former part but often then erroneously showed some kind of catalysed reaction with a lower activation energy.

(e)(i) This question was not generally well answered, as candidates were not able to associate electron movement with the correct stage of the reaction and often were very imprecise with their positioning of curly arrows. Many answers had the arrowhead and tail reversed, with a lot of responses suggesting electron movement *from* extra H^+ species.

(ii) This question was reasonably well answered.

(iii) This question was generally well answered, although stray vowels or unspecific/incorrect numbering were observed. Candidates are reminded that the longest carbon chain is the important root of a molecule's name.

(f) (i) Candidates were able to provide a correct answer here by analogy of the 2-bromobutylium species on the left-hand side.

(ii) This was found to be a demanding question, although candidates who recognised the necessary reference to the stability of the intermediate were able to give a reasonably straightforward AO1 account for full credit.

1(a)		**2**

Mg^{2+}	24	12	10
Al^{3+}	27	13	10

4 correct = 2 marks
2 or 3 correct = 1 mark
0 or 1 correct = 0 marks

1(e)(i)		**2**

Na	solid	10–14
S	gas	0–4

4 correct = 2 marks
2 or 3 correct = 1 mark
0 or 1 correct = 0 marks

1(e)(ii)	acid–base **OR** neutralisation	**1**

Question 1

(a) The majority of answers were correct.

(b) Some excellent explanations were seen. Occasionally the aluminium ion was incorrectly described as larger than the magnesium ion. In general, the explanations focussed, at least in part, on the difference in nuclear attraction which arises as a result of the difference in nuclear charge and the same shielding effect of both species.

(c) Clearly labelled diagrams, which showed a regular arrangement of sodium cations surrounded by delocalised electrons, were seen in a significant proportion of answers. Poorer diagrams showed randomly placed positive ions or made no reference to 'delocalised' electrons. Occasionally, the positive symbols were described incorrectly as 'nuclei' or 'protons' or 'atoms'.

(d) (i) The majority identified the structure and bonding present in silicon. A much smaller proportion used this knowledge to explain why silicon had a high melting point in terms of the high energy requirement required to break many strong covalent bonds. A common misconception described intermolecular forces in a giant molecular structure.

(ii) Of those that annotated Fig.1.1, many correctly compared the melting points of P, S and C*l* with those of the three metals provided. Comparison of the melting points between P, S and C*l* was less well known.

(e) (i) This question proved to be demanding. The state of both sodium oxide and sulfur dioxide under room conditions was not well known. Misunderstanding of the term 'state of oxide' and oxidation number was seen on occasions. A common misconception described aqueous solutions of sodium oxide with a pH of 7.

(ii) Correct identification of this reaction as acid–base or neutralisation was common.

(iii) The product for the reaction of P_4O_{10} with an excess of water was well known. Equations were not always balanced.

(f) (i) The majority of answers demonstrated a good understanding of the term amphoteric.

(ii) There was some confusion with this reaction and the reaction of aluminium oxide and sodium hydroxide. The correct formula of the product was not well known.

3(a)	**M1** rates of forward and reverse / backward reactions are equal **M2** no change in measurable properties **OR** concentration of reactants **AND** products remain constant	**2**

3(b)(i)	**M1** change to appearance of the mixture: (goes) darker red / more red **M2** change to relative concentration of $Fe^{3+}(aq)$: decreases **M3** change to the value of equilibrium constant, K_c: constant / none	**3**

3(b)(ii)	Method 1 **M1** conc of initial conc of Fe^{3+} and SCN^- = 0.0020 (mol dm^{-3}) **M2** at equilibrium the conc of Fe^{3+} and SCN^- are (M1 – 0.000423) **M3** use 0.000423 ÷ (**M1** – 0.000423)^2 to calculate K_c = 170(.089) **M4** units = $mol^{-1}\ dm^3$ **OR** $dm^3\ mol^{-1}$ Method 2 **M1** no mol $FeSCN^{2+}$ = $4.23 \times 10^{-4} \times 25 / 1000 = 1.0575 \times 10^{-5}$ **M2** no mol Fe^{3+} and SCN^- are (5×10^{-5} – **M1**) = 3.9425×10^{-5} **M3** 0.000423 ÷ (**M2** × 1000 / 25)^2 = 170(.089) **M4** units = $mol^{-1}\ dm^3$ **OR** $dm^3\ mol^{-1}$	**4**

Question 3

(a) Accurate descriptions of this definition were seen. Some weak responses were vague and incomplete, e.g., 'when forward and backward reaction is equal'. The misconception that dynamic equilibrium occurs only when 'concentration of reactants and products are the same' was also seen.

(b) (i) This question proved to be challenging. Many candidates were unable to apply le Chatelier's principle to this novel equilibrium reaction.

(ii) Some excellent answers, presented clearly and logically, were seen in this demanding question. A significant proportion of candidates confused the terms 'amounts' and 'concentrations'. Understanding of the different mathematical relationship between 2x and x^2 was not always appreciated. Correct deduction of the unit was common.

(c) Many answers were not based on the knowledge that 4s electrons are lost before 3d electrons when the electronic configuration of a transition metal cation is described.

(d) Correct dot-and-cross diagrams were seen. Weaker responses showed correct bonding electron arrangement and ignored the additional electron present in a SCN^- anion.

2(a)(i)	hydrolysis	**1**
2(c)(ii)	$Mg_3N_2 + 6H_2O \rightarrow 3Mg(OH)_2 + 2NH_3$	**1**
2(c)(iii)	**M1** due to $OH^-(aq)$ **M2 EITHER** acid-base reaction of water by ammonia **OR** dissociation of (some) $Mg(OH)_2$	**2**

Question 2

(a) (i) This question was well answered.

(ii) The majority of answers gave a formula for the different Period 3 element when the question asked for a name. Common errors included phosphorus chloride, omitting the oxidation number, and phosphorus pentachloride, which is a solid.

(b) (i) The dot-and-cross diagram of SCl_2 was well answered. A common error included annotating the electrons around the S atom as 'x' in contradiction of the rubric.

(ii) The VSEPR theory was not well known when applied to a central S atom with two bonding pairs and two lone pairs of electrons in the outer shell. The majority of candidates focused on the two bonding pairs in SCl_2, stating that the shape was linear with a bond angle of 180°.

(c) (i) This question was well answered.

(ii) Most candidates gave a correct equation for the reaction of magnesium nitride with water. The most common error was an incorrect formula for magnesium nitride and magnesium hydroxide given as MgOH or MgO.

(iii) The marks were awarded to a minority of candidates. Very few candidates stated that hydroxide ions were present either from the dissociation of $Mg(OH)_2$ in solution, or from a reaction between ammonia and water.

(d) (i) The empirical formula of boron nitride was well answered with a common error of Br used to represent boron.

(ii) Many candidates gave the correct answer of graphite that has a structure similar to that of boron nitride. SiO_2 and graphene were common incorrect answers.

3(a)	**M1** (if a change in conditions occurs) the equilibrium shifts **M2** to minimise the change in conditions	**2**

3(b)	**M1** change in appearance paler red / more yellow / less orange **M2** change in relative concentration of $FeSCN^{2+}(aq)$ lower **M3** change in value of the equilibrium constant, K_c lower	**3**

3(c)(i)	**M1** *deduce the concentration of SCN^- present in initial mixture* [SCN^- eq'm] = [Fe^{3+} eq'm] = [original SCN^-] – [$FeSCN^{2+}$ produced] $1.30 \times 10^{-3} + 0.300 \times 10^{-3} = 1.60 \times 10^{-3}$ mol dm^{-3} **M2** *find initial amount of Fe^{3+} initially* M1 $\times$ 25 / 1000 = amount of Fe^{3+} added initially = 4.00×10^{-5} (mol)	**2**

3(c)(ii)	**M1** value for K_c = 178 **M2** units = mol^{-1} dm^3	**2**

Question 3

(a) Defining Le Chatelier's principle proved challenging for some candidates with a common answer being a definition of a dynamic equilibrium, in terms of the rates of forward and backward reactions. A further error referred to a shifting of the system rather than of the equilibrium, following a change in conditions.

(b) Many candidates were able to deduce the directed changes to an equilibrium subjected to a change in temperature. The most common incorrect answer was 'no change' for the effect of temperature change on the numerical value of K_c, the equilibrium constant.

(c) (i) A minority of candidates were able to calculate the initial moles of $Fe^{3+}(aq)$ in an equilibrium mixture: $Fe^{3+}(aq) + SCN^{-}(aq) \rightleftharpoons FeSCN^{2+}(aq)$, where the concentrations of two species in the equilibrium mixture were provided in the question.

Some candidates correctly calculated the moles of $SCN^{-}(aq)$ and $FeSCN^{2+}(aq)$ at equilibrium, but then proceeded to subtract these two values instead of adding them, to give the initial moles of $Fe^{3+}(aq)$.

(ii) Many candidates were unable to calculate a correct value for the K_c, for this reaction, but gained credit for the correct units.

2(a)	curved line from (0,0) to reach same horizontal line	**1**

2(b)	equilibrium position would / it moves to left / reactants	**1**
	forward reaction is exothermic (so disfavoured by increase in *T*)	**1**

2(c)	mole fraction NO $\frac{0.605}{1.60}$ (= 0.378125) NO_2 $\frac{0.605}{1.60}$ = (= 0.378125) N_2O_3 $\frac{0.390}{1.60}$ (= 0.24375)	**1**
	(partial pressure / kPa 38.19 38.19 24.62) $K_p = 24.62 \div 38.19^2 = 0.0169$	**1**
	kPa^{-1}	**1**

2(f)(i)	magnesium hydroxide / $Mg(OH)_2$	**1**

Question 2

(a) Some candidates appeared to confuse graphs of rate versus time with those showing concentration versus time.

(b) Well-rehearsed arguments related the significance of increasing temperature on the position of equilibrium to the example given. More able candidates showed no ambiguity and made explicit reference to the effect of an increase in temperature on the position of this equilibrium in terms of either the '*forward reaction is exothermic'* or '*the reverse reaction is endothermic'*. Less precise answers included '*the reaction moves backwards as it's an exothermic reaction'*.

(c) This question proved to be demanding. A few candidates found K_p using an appropriate method; they worked out the mole fraction of the components, converted them into partial pressures and then substituted these values into the correct expression for K_p. Weaker candidates calculated K_p directly from moles and included moles in the units. Deduction of units also presented difficulties. Many answers gave units in terms of Pa with no attempt to convert kPa to Pa within the calculation.

(d) Lightning was well known as a natural process responsible for formation of atmospheric NO and NO_2. 'Catalytic converter' as the man-made process was a common incorrect answer.

(e) (i) Many correct definitions of the term free radical were seen. Ambiguous descriptions referred to '*unpaired electron'* as '*unshared*' or '*unbonded*' or '*free*'. Common incorrect answers referred to a free radical as *'a species with a lone pair of electrons'*.

(ii) The pair of equations which showed the catalytic role of NO_2 in the oxidation of SO_2 were not well known. The equation to represent the regeneration of NO_2 appeared to cause most difficulties with many incorrectly balanced equations seen.

(iii) The environmental consequence of the oxidation of sulfur dioxide was well known.

(f) (i) Identification of a magnesium containing base was relatively common. Incorrect answers included salts, rather than bases, which contained magnesium or bases that were not made from magnesium. The formula of the compound identified was not always correct.

(ii) This equation proved challenging for many. Some answers incorrectly balanced equations which included all correct species. Use of an incorrect formula for magnesium nitrate was a common error, including Mg_3N_2.

(iii) Identification of the change in thermal stability down Group 2 nitrates was fairly well known.

3(a)(i)	$4P / P_4 + 5O_2 \rightarrow P_4O_{10}$	**1**
	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	**1**

3(a)(ii)	proton / H^+ donor	**1**
	partially dissociates (in solution / in water / when aqueous)	**1**

Question 3

(a) (i) Fully balanced equations were seen by more able candidates.

(ii) This definition was described accurately in most candidates answers. A common misconception was seen in weaker responses that described a weak acid as one that *'did not completely dissolve'*.

(b) (i) Many candidates gave a definition of the enthalpy change which occurs when 'one mole of product is formed from its elements'. It was less common for the elements to be described as 'in their standard states'.

(ii) More able candidates gave excellent and logical workings that contained clearly labelled energy cycles. It was common to see an expression for the different bond energies added or subtracted from each other, in a variety of possible incorrect combinations, with no attempt to show how these expressions were deduced. The stoichiometry of the equation was ignored by some.

(iii) Excellent answers were seen which identified phosphorus in H_3PO_3 as both oxidised and reduced and linked the process of oxidation and reduction to the relevant changes in oxidation number for the three phosphorus containing species. Weaker responses were less specific, for example '*H_3PO_3 is oxidised and reduced'*. Oxidation numbers were sometimes described incorrectly.

(c) (i) Some candidates identified **A** as steam or $H_2O(g)$, other candidates were less precise and identified **A** as water. A common incorrect answer was NaOH.

(ii) Many unambiguous representations of 2-bromopropane were seen. Occasionally the central carbon atom was drawn with a valency of 3.

(iii) Description of this esterification reaction as a condensation reaction was relatively well known. Incorrect answers included hydrolysis and oxidation.

(iv) This question proved to be demanding. Candidates were expected to ignore all bonds which were common to both reactant and product. While the appropriate bond, C=O, was identified in many answers C-O and C-H were also seen. Most candidates only identified a range of wavenumbers where a change in absorption occurs rather than described if the absorption appeared, disappeared or moved.

(d) (i) More able candidates deduced a formula for the correct organic product in the equation. The inclusion of H_2O as the additional product was less common, even when included the equation was not always balanced.

(ii) The appropriate formula to deduce the number of carbon atoms present in **T** was not well known and many confused calculations were seen. It was common for candidates to give a structural formula for **T** rather than the required molecular formula.

3(a)	**M1** (energy change) when 1 mole of water is formed **M2** from an (aqueous) acid and an alkali / aqueous base	**2**

3(c)(i)	$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$	**1**

3(c)(ii)	(additional) energy is involved / required to fully ionise / dissociate (weak / ethanoic acid) **OR** (additional) energy is involved / required to donate/ release H^+ ions	**1**

Question 3

(a) Many candidates incorrectly stated, 'enthalpy change when 1 mole of acid reacted with 1 mole of base'.

(b) (i) Many candidates calculated the initial amount of H_2SO_4 instead of calculating the amount of acid that was neutralised by the NaOH.

(ii) Very few candidates deduced that Q = 5710 J and the mass of the solution was often incorrectly given as either 1.00 g or a number derived from calculating the masses of NaOH and H_2SO_4 in the reaction.

(c) (i) Some good answers were seen.

(ii) Several candidates knew that CH_3COOH was not fully dissociated, as it is a weak acid, compared to the strong acid HC*l*, which resulted in a less negative value for the $\Delta H_{neut.}$ Very few candidates knew that the difference in values of the respective ΔH_{neut} values was due to the additional energy required to fully dissociate the weak acid such that less energy is released after neutralisation.

4(a)(i)	$2KCl + H_2SO_4 \rightarrow K_2SO_4 + 2HCl$ **OR** $KCl + H_2SO_4 \rightarrow KHSO_4 + HCl$	**1**
4(b)(i)	(from colourless) to purple	**1**
4(b)(ii)	$K_C = [H_2][I_2] / [HI]^2$ **OR** $0.0217 = [H_2][I_2] / [HI]^2$	**1**
4(b)(iii)	**M1** [HI] at equilibrium = 1.7 / 2 = 0.85 **M2** idea that $[H_2] = [I_2]$ in equilibrium mixture **M3** correct calculation based on correct rearrangement of equation in (b)(ii), 0.0217 and their M1 = 0.125	**3**
4(c)(i)	new mixture will have a lower / smaller conc / amount of H_2 and I_2	**1**
4(c)(ii)	endothermic **AND** decrease in temperature causes position of equilibrium to shift in exothermic direction	**1**

Question 4

(a) (i) Some good answers were seen.

(ii) Very few candidates knew that hydrogen iodide is not prepared in the same way as hydrogen chloride because the hydrogen iodide and concentrated H_2SO_4 would undergo a redox reaction to form a variety of products to include SO_2, H_2S and I_2.

(b) (i) A small number of candidates recognised that when a sample of colourless HI(g) is placed in a sealed container and allowed to reach equilibrium the colour changes to purple, attributed to the formation of I_2(g).

(ii) This question was well answered by many candidates.

(iii) The calculation of the concentration of iodine in an equilibrium mixture was generally not well answered. Candidates who deduced that [HI] is (1.70 ÷ 2 =) 0.85 mol in the equilibrium mixture tended to gain some credit. Very few used the expression $\sqrt{(0.0217 \times 0.85)}$ to calculate the $[I_2]$ as 0.125 mol dm^{-3}.

(c) (i) Some candidates were able to describe the difference in composition of the equilibrium mixture at the two temperatures.

(ii) Some candidates deduced that the reaction is endothermic in the forward direction leading to a lower concentration of H_2 and I_2 in the equilibrium mixture.

4(a)	Method 1 • identification of number and type of bonds when all are broken 2O-H + 4C-H + C=C **OR** (2 × 460) + (4 × 410) + 610 • Identification of number and type of bonds when all are made O-H + 5C-H + C-C + C-O **OR** (460 + (5 × 410) + 350 + 360) • calculation of energy to break bonds – energy to make bonds 3170 – 3220 = – 50 (kJ mol^{-1}) Two correct for one mark, three correct for two marks. **OR** Method 2 • identification of number and type of relevant bonds broken (including the new C-H only) 2O-H + C=C **OR** (2 × 460) + 610 • identification of number and type of relevant bonds made (including the new C-H only) O-H + C-H + C-C + C-O **OR** (460 + 410 + 350 + 360) • calculation using energy to break bonds – energy to make bonds 1530 – 1580 = –50 (kJ mol^{-1}) Two correct for one mark, three correct for two marks.	**2**
4(b)	**M1** amount ethanol increases **M2** equilibrium moves to right **AND** to produce fewer moles gas OR smaller amount of gas (to reduce the pressure)	**2**
4(c)(i)	acid **AND** it donates a proton / H^+	**1**

Question 4

(a) Stronger candidates showed clear logical working and selection of appropriate data. Common errors included mis-reading the table, using an inappropriate value for a correctly identified bond, or ignoring the C-O bond during the reaction.

(b) Reference to the effect of increasing pressure on the amount of ethanol produced at equilibrium was not always included. Many explanations were given in terms of the change producing a shift to the position of equilibrium to the right; the significance of producing a smaller amount of gas to explain this change was mentioned less frequently.

(c) (i) Many candidates were able to deduce the behaviour of H_3PO_4 as an acid or an electrophile. Some explanations were not specific to step 1 in the reaction.

(ii) Some candidates were able to identify the correct species in step 2. Explanations were often vague, for example, 'seeking electrons' or 'being attacked by a pair of electrons' rather than 'accepting a lone pair'.

(iii) A small proportion of candidates applied knowledge of the drawing of mechanism arrows to represent electron movement when bonds are made and broken in this unfamiliar context. A common error showed arrows pointing in the opposite direction to the way an electron pair moves when a bond is made or broken. Sometimes an arrow did not start from the bond being broken or from the lone pair of electrons on an atom that is used to make a specific bond.

(iv) It was generally well known how a catalyst affects a reaction. Some ambiguity was seen in answers that described the effect of catalysts as increasing the number of collisions between particles rather than increasing the frequency of successful collisions.

(v) Stronger candidates clearly described the regeneration of the catalyst. Some candidates were vague and imprecise, for example 'the catalyst did not change in the reaction'.

(vi) Some candidates showed an excellent understanding of the consequence of more alkyl groups present on the C^+ of the intermediate in terms of a larger inductive effect. Some candidates incorrectly referred to the inductive effect of propan-2-ol rather than the carbocation intermediate which was responsible for the production of propan-2-ol.

(d) The sideways overlap of two p orbitals to form a pi bond between the two carbon atoms was well known. Descriptions of the formation of the sigma bond by direct overlap of sp^2 hybrid orbitals were less common.

3(a)(i)	(a reaction where) two (or more) compounds / reagents / molecules form (only) **one product**	**1**

3(d)(i)	increases the rate of reaction	**1**
	by providing an alternative reaction pathway of lower E_a	**1**

3(d)(ii)	equilibrium moves to right/products where fewer moles / smaller amount of **gas**	**1**

Question 3

This question was largely framed around organic chemistry, with the inclusion of some physical and stoichiometric work. It was well attempted on the whole, although knowledge of some key definitions was seen to be weaker.

(a) (i) Many candidates appreciated that a single product was the outcome of an addition reaction, but many others described/attempted to describe an addition reaction without indicating that only one product was formed.

(ii) This was well answered by most candidates.

(iii) The major errors here were butan-1-one, butan-1-ol/butanol or propanal/propan-1-al. Candidates are reminded that there is no need to offer a locational number for the aldehyde group.

(iv) Very few candidates converted the quantities to moles, so simply offered 96% of 5000 kg as the answer.

(b) (i) Candidates seemed largely unaware of the nucleophilic addition mechanism. Common errors and omissions included:

- incorrect placement of the lone pair of electrons on the N of the CN^- ion.
- giving the intermediate a positive charge which was subsequently attacked by the CN^- nucleophile.
- mechanisms involving the H (of HCN) / H^+ attacking the intermediate negative ions.

- curly arrows inaccurately coming from the C (in the C=O group).
- the H (of HCN) shown attacking the O (in the C=O group) to form an intermediate hydroxyl derivative with a positive charge.

(ii) The first and third marks of this question gave the largest number of variants, with Fehling's reagent or DNPH offered for experiment 1. For experiment 3, many answers erroneously gave a change or loss of colour rather than a method of noting that no reaction had occurred.

(iii) There were some very good answers here, although some increased the number of H atoms in the product and/or H_2 appeared as a product.

(iv) This was generally very well answered, with occasionally some candidates circling two C atoms or the entirety of HOCHCN.

(c) It appeared that many candidates did not understand the term molecular formula and gave structural formulae throughout.

(d) (i) Some answers referred to a change in position of the equilibrium/yield, which was not credited.

(ii) Many answers excluded the word gas and just stated 'less/fewer (number of) moles'. Although the reaction in question was entirely in the gas phase, precision in an answer is crucial. A few candidates gave the incorrect direction of the equilibrium shift.

(iii) Many candidates used a value of 436 in their answers, which suggests the use of a Data Booklet. Centres are reminded that Data Booklets are **not** to be issued to candidates for these papers.

(iv) Any mention of hydrogen bonding was not credited, in accordance with the convention and definition used by this qualification.